

Team 17
Preliminary Design
Review

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1. Introduction and Summary

1.1. Team Introduction

Andrew Mills University: Cal State Long Beach; Long Beach, CA	• 8 week Boeing summer internship • StellarXplorers 2 year member and team leader • 2 year robotics member • 4 years of high school engineering courses
Anny Antunovich University: Cal State Fullerton; Fullerton, CA	• Helpful • Attentive • Cooperative
Anthony Garcia University: Cal State Northridge; Northridge, CA	• Team player • Approachable • Flexible
Erik Jensen University: Cal State Northridge; Northridge, CA	• Robotics knowledge from 4 years in school clubs • Learned to use Solidworks and AutoCAD from college class • Participated in a week-long NCAS(NASA Community College Aerospace Scholars) program in Palmdale, CA
Jacob Powell University: Cal State Fullerton; Fullerton, CA	• Full time Software Engineer proficient in C++, Python, and JavaScript • 1 year Robotics experience • Project Manager at the CSUF branch of ACM

	(Association for Computing Machinery)
Keir Bildgen University: Cal State Long Beach; Long Beach, CA	• Proficient in SolidWorks • Two years of VEX Robotics experience
Marco Marrufo University: Cal State Long Beach; Long Beach, CA	• NASA Ames Machine Learning/Software Engineering Intern • NASA Armstrong Electrical Engineering Intern • NASA Glenn Software Engineering Intern • NASA Armstrong Pathways Civil Servant • CSULB Beach Launch Team
Ryan Luong University: Cal State Fullerton; Fullerton, CA	• Currently in Titan Rover • Worked on the mobility system for the rover • Currently making the UI for the rover

1.2. Mission Overview

1.2.1. Mission Statement

Team 17 hopes to land a lander on Vesta and collect data to understand what it means for an asteroid to be considered a protoplanet and understand how asteroids were able to hold water in the past. The data collected will be Vesta's rocks. The rocks will be collected from areas where there are high levels of water evaporation. By collecting this data, Vesta's rocks will help us understand what is a protoplanet and help us understand how other

asteroids and planets were able to hold water despite water evaporating in the air.

1.2.2. Mission Requirements

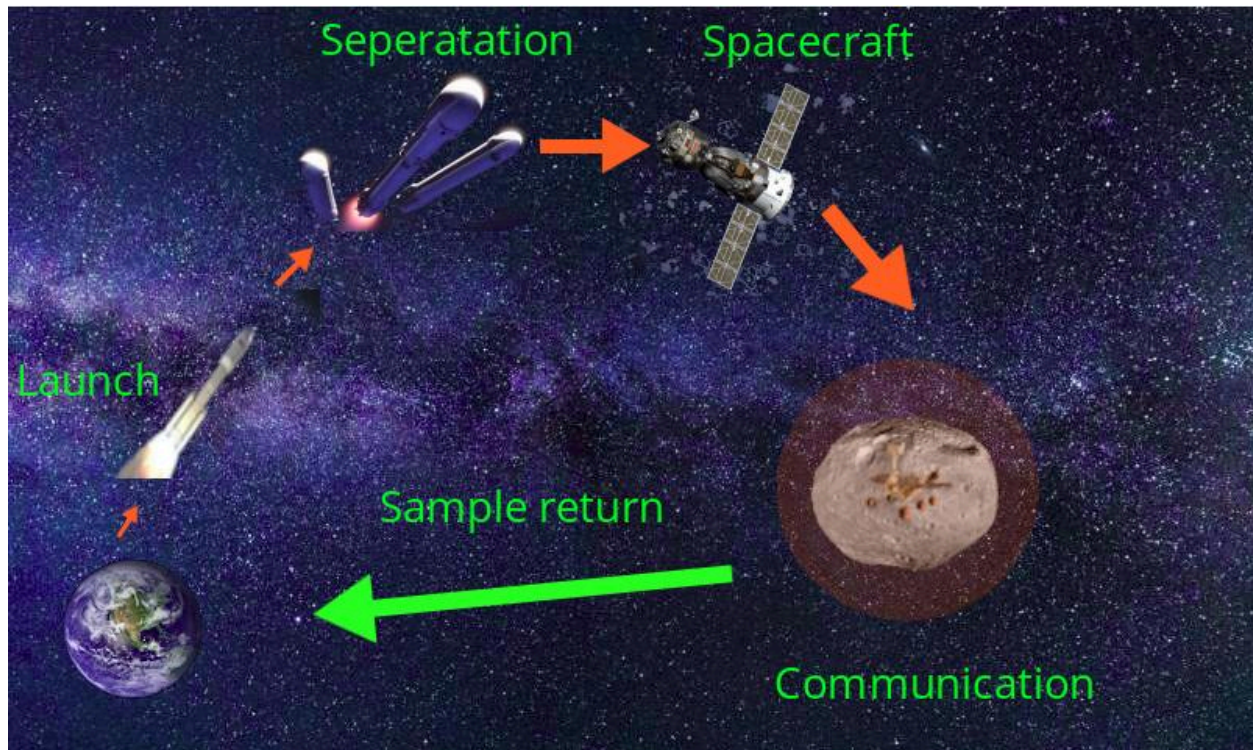
The mission requires that the spacecraft must have the mass constraint up to 15kg and must be able to fit inside of a 50cm cube. The spacecraft must have a mobility system that will visit three sites on Vesta and must travel a total of 150 meters. It must also operate at least 72 hours. The spacecraft must also have a communication instrument that will communicate with the orbiter. Lastly, the budget must not exceed over 45 million dollars.

1.2.3. Mission Success Criteria

For the mission to be considered a success, evaluating the parameters of a protoplanet and then comparing them to the current condition of Vesta is needed. With a little more thorough research, there is a possibility that Vesta could overcome its limitations to obtain the protoplanet title.

The next goal is to accomplish in obtaining a more aged sample of a substance that contains water. If this sample is to be preserved, it can then be evaluated to determine the age of the rock and later try to assess a trajectory path from which the collision had occurred. If this can be identified, then assessed a more scientific approach as to where another source of water originated, then the possibilities of another habitable planet is within reach.

1.2.4. Concept of Operations (Graphic)



The path to Vesta is listed as so. The spacecraft will launch towards space and separate to reveal it. As it reaches near Vesta, the descent system will land a rover on Vesta, and the spacecraft will be circling Vesta for communication. While the rover is in Vesta, it will send back data to the spacecraft to Earth. Once the science objective is done, the rover will return to Earth with the Vesta's sample through the spacecraft.

1.2.5. Major Milestones Schedule

During Phase A, the team will research potential lander designs, Vesta's conditions, brainstorm what needs to be done on Vesta, and understand what everyone's role is on the team. Phase B will be conducted once these four activities are completed. Phase B consists of listing the requirements and eliminating potential lander designs that do not meet the criteria. Scientific instruments are picked to

satisfy the mission's objective successfully. Phase C and Phase D consist of the preliminary design and critical design review, which criticism will be taken seriously for the whole mission to be successful. After criticism is taken into account, the team will move to Phase E, in which the launch and the mission will be happening in this phase.

1.3. Descent and Lander Summary

The rover currently has a height of 20 cm with the length and width of 20 cm. It is a rectangular shape that weighs about 376 pounds. The rover will also have two wheels on each side. It will have a crane in the front for the rock abrasion tool. The rover will be deployed using a heat shield, parachute, and lastly, descending on airbags.

1.4. Payload and Science Summary

The rover houses four instruments pertaining to the science objective: the ultraviolet imaging spectrometer, the rock abrasion tool, microscopic imager, and the radioisotope heater unit. The ultraviolet imaging spectrometer detects particles through ultraviolet light, the rock abrasion tool drills, the microscopic imager allows a closeup view of anything, and the radioisotope heater unit provides heat through radioactive decay. The radioisotope heater unit will be inside of the rover. The rock abrasion tool will be housed on a crane with the ultraviolet imaging spectrometer and the microscopic by the rock abrasion tool side.

2. Evolution of Project

2.1. Evolution of Descent and Lander

Initially, the plans of making the rover consist of a height of forty cm, length, and width of forty cm fell apart when taken account of Vesta's temperature. Because it is much bigger with a 40 cm all around, the heat particles

spread rather than compress. Thus, with that in mind and Vesta's temperature, it would have taken a long time to heat the rover, which is not the step the team wants when the main goal is to heat the rover as quickly as possible. Secondly, the plans to make it the height of 10 cm was thought of, but it comes to another issue that it might not fit all the scientific instruments. Some might possess higher than 10 cm. But it solves the problem of heating because it is much smaller. Thus, the plans of making the rover a height of 20 cm were decided. It allows ample height room for the instruments and still has somewhat the heating speed of the previous design. Lastly, the model also has a crane.

2.2. Evolution of Payload

With the goal of heating the rover, the idea was to group the scientific instruments as close as possible was the original plan. But that doesn't help heating the rover entirely, and it would just create a heating aura around the scientific instruments. Also, the heat coming out of the devices is not enough to combat Vesta's temperature. Thus, with the addition of the radioisotope heater units, the plans of putting them in the corners, as well as one in the middle, was decided. The radioisotope heater units are heating the entire rover as well as keeping the instruments warm as it covers all the bases. With this in mind, the scientific instruments should be in the area of the radioisotope heater units and other scientific instruments so that it is intaken heat from itself, other instruments, and the radioisotope heater units.

2.3. Evolution of Mission Experiment Implementation Plan

Initially, the ultraviolet spectrometer, the rock abrasion tool, and the microscopic imager were the main three that were part of the science objectives. However, the addition of the radioisotope heater unit was crucial as the team overlooked Vesta's temperature. Additionally, cameras were added. While this does not affect the science objective, cameras

were added to provide the rover's point of view. The mission criteria never change within the project, but instrumentation changed.

3. Descent and Lander Design

3.1. Selection, Design, and Verification

3.1.1. System Overview

The rover has a height of 20 cm, length and width of 40 cm, and is a rectangular shape. It will also have a crane to house the rock abrasion tool, the ultraviolet spectrometer, and the microscopic imager. The decision to make it a 40 by 40 by 20 cm is partly due to heating. A larger design, such as a 40 by 40 by 40 cm, will make the heat particles spread instead of compressing. The goal of the model is to heat the rover quickly the moment it lands on Vesta. The rover will still heat up with a 40 by 40 by 40 cm, but it would have taken more time to do so. Also, the current design of the rover allows ample room for scientific instruments.

The rover's descent system is composed of a heat shield, parachute, and airbags. There will be eleven parachutes, and it will be round and large. The airbags used in this mission will be coming from the NASA Pathfinder mission.

3.1.2. Subsystem Overview

The decision to pick three large round parachutes has to do with the fact of the success of the Orion Parachute. Instead of creating parachutes specifically for the rover, it would be much more efficient to look at other parachute design and modify through there. Because there are eleven parachutes, if one of them fails, then the parachute system will still operate because there will be ten more. Also, Orion Parachute was used successfully, so there's an added security that it works. The airbags were picked specifically because other designs the team looked at were specifically targeted for larger rovers. The rover for this mission is not quite the size of the larger rovers. Also, the designs to land the larger rovers

exceed the requirements. The airbags do not possess any of these issues.

3.1.3. Dimensioned CAD Drawing of Entire Assembly

The design of this current rover allows the heat particles to compress rather than spread. The crane is included specifically for the rock abrasion tool, the ultraviolet spectrometer, and the microscopic imager. By putting those three together, it would save cost as a rotating system would not be needed to use the instruments individually. Also, if a rotating system was to be implemented, there is an issue that it can malfunction or damage during the lander or during the mission. If it is damaged or malfunction, then the instruments can not do its job regardless if it works or not. The possibility to migrate this is to move the rover in the direction of whatever devices are facing, but that adds another issue that the rover may not align the drilled area correctly.

3.1.4. Manufacturing and Testing Plans

The manufacturing process will be outsourced to the Xometry manufacturing site, who will CNC the part with series 5083 Aluminum. This series of Aluminum was chosen for its structural integrity at the low temperatures of Vesta. According to Xometry, the required man-hours would be 14 workdays.

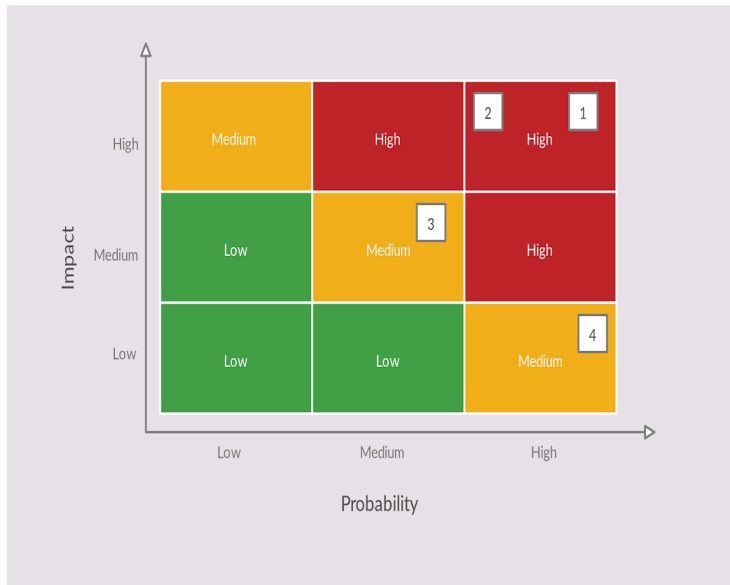
3.1.5. Validation and Verification Plans

To test the rover's design and materials, it must be tested in a cold and isolated room in the laboratory. Because of how cold Vesta is, testing in a place that simulates the chilly and remote environment, the rover will give the team an idea of whether or not the design is suitable for Vesta. The distribution of the weight of the

rover must be tested as well. Because of the bumpiness of Vesta, the rover will be moving side to side. Examining the distribution of the weight also means considering the shape of the rover. Testing the distribution, a location that is bumpy, and finding a form that distributes the weight evenly will valid the design of the rover. Given that the rover will be unseeable from Earth aside from the camera feeds, having a system that will verify that the design is working from millions of miles away is essential. Fortunately, because the rover will be sending data from Vesta as well as having the camera feeds, those two alone will verify if it is still working.

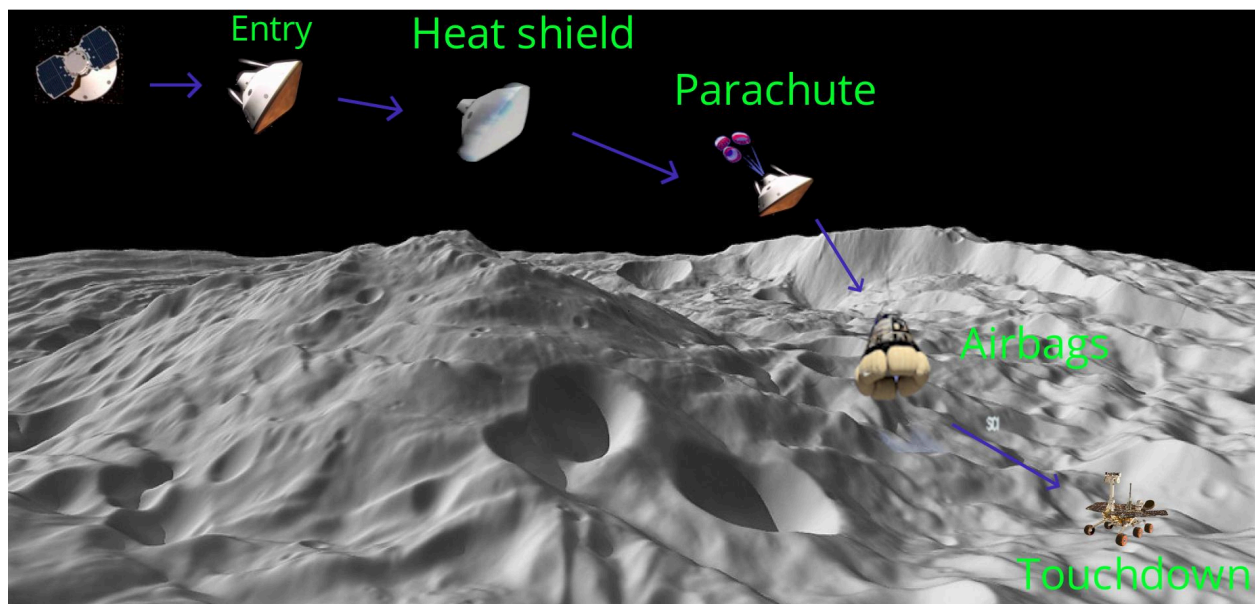
3.1.6. FMEA and Risk Mitigation

Functions	Failure mode	Effect	Sev	Cause	OCC	Prevention	Detection	DET	RPN	Recommendation Action
To provide a mobility system	The entire rover fall over	The entire mission fails	10	The weight of the rover is not distributed properly	1	Distribute the weight equally	Design the rover where the weight will distribute equally	4	40	The design should be checked if the weight is distributed equally
To house the instruments	The instruments stop working	The objective can't be satisfy	10	Vesta's cold temperature	10	Keep the rover warm throughout the whole mission	The instruments are damaged	5	500	Look back on past NASA's mission to figure out how to keep the rover warm
To house the communication instrument	The cameras and the data is no longer being sent to Earth	The point of view of the rover and information is gone	8	The distance between the satellite and the rover is too far	7	Place the spaceship at an area where it is around the location the rover will visit	Information is no longer being sent to Earth	10	560	Test the maximum distance of the communication instrument



Rank	Approach	Risk Title
1	Research	Rover falling over
2	Research	Instruments stop working
3	Mitigate	Cameras stop displaying
4	Watch	Going over the budget

3.1.7. Performance Characteristics and Predictions



The mission criteria is to collect a Vesta's sample and return it to Earth. To start, Vesta is very cold. So the cold weather of Vesta could affect the descent system. The coldness could affect wind speed. The parachute could be unstable if the wind

speed goes faster. The airbags could not land near the area it would visit if the wind is strong enough to push it around. To mitigate this, it would be better to include a much larger battery and to have the rover travel to where it is at the location it should be traveling. The wind speed is a variable that is outside of the descent system control. The angle where the rover will be released should also be taken into consideration. The angle should be within the areas which the rover will be visiting.

3.1.8. Confidence and maturity of Design

To test the design of the rover, placing the rover in a cold and isolated room and remain in place will prove if the materials of the rover will remain intact during the duration of the mission. Secondly, testing the heating of the rover will also happen in the room. The heat must be able to keep the instruments running and prevent the cold from affecting it. Next, testing the rover in a bumpy location will determine if the design is suitable for Vesta. The rover will be driving the whole time. While this is happening, the rover will be sending false information to the team to test the communication of the rover. The rover will be driving far enough to simulate like Vesta, where the team is unable to see the rover. If the rover is unsuitable for both the cold and bumpy environment testing, the design of the rover must be changed.

3.2. Recovery/Redundancy System

It is possible that the ultraviolet spectrometer could be damaged during the duration of the rover. Losing the ultraviolet spectrometer will make but not impossible for the mission to be accomplished. Losing the ultraviolet spectrometer means losing the ability to determine if there are water particles in the rocks. To mitigate this issue, the rover must rely entirely on the rock abrasion tool and the microscopic imager to determine if there is water in the rocks. The microscopic imager provides an up-close view of the rocks due to its built-in microscope. While this does not have the same ability to detect water particles like the ultraviolet spectrometer, the ability to look up close to the rocks is the backup

plan for the malfunction ultraviolet spectrometer. The loss of communication is another issue that must be looked at. In the case of communication loss, there will be attempts to reconnect the rover and the communication to Earth.

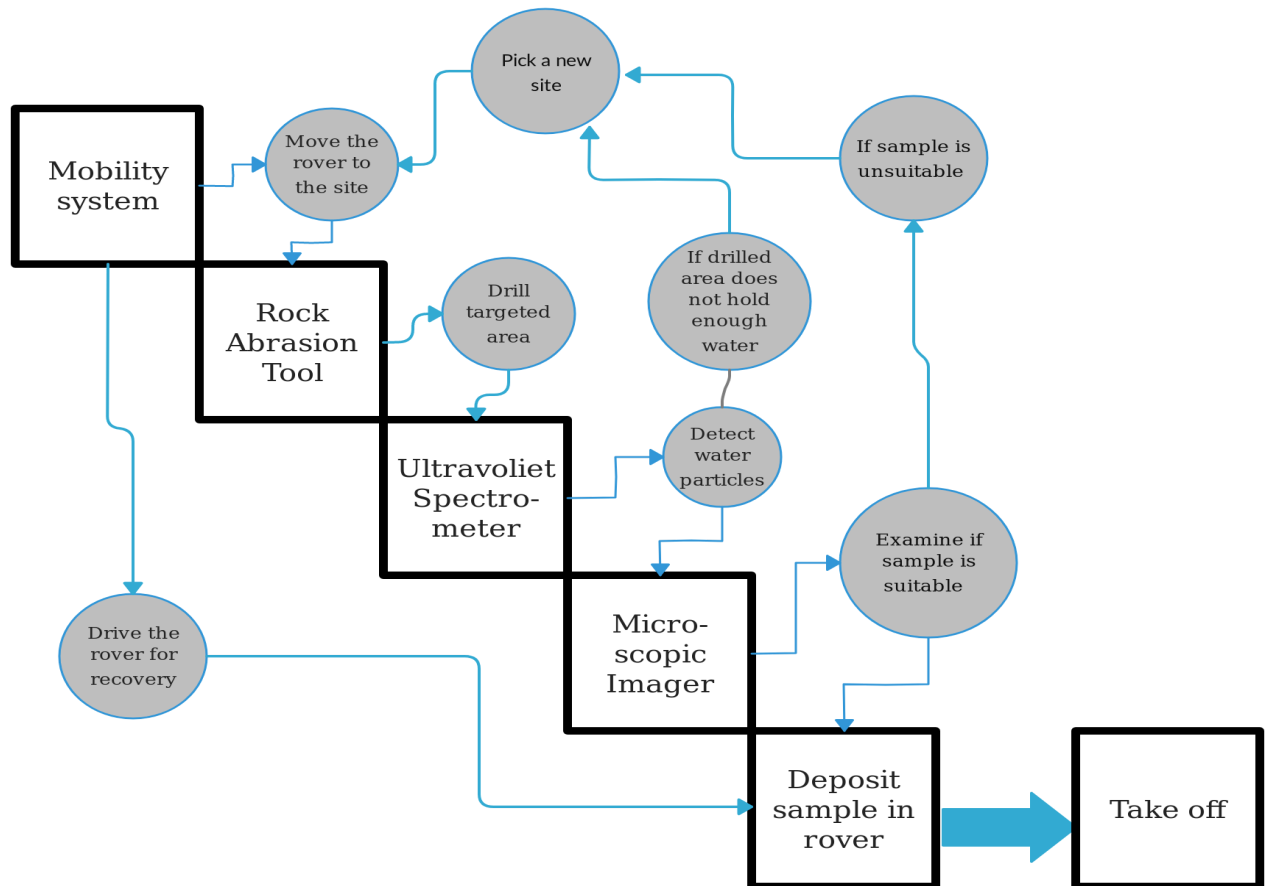
3.3. Payload Integration

Inside of the rover, the radioisotope heater units will be in the corners of the rover and in the middle. This interaction will heat the other instruments. Placing the devices around where the radioisotope heater unit will be will warm the other instruments. The combination of the heat from the instruments and the radioisotope units will radiate a big aura of warmth. This will make the rover warm. The crane provided allows the ultraviolet spectrometer, the rock abrasion tool, and the microscopic imager to be right next to each other.

4. Payload Design and Science Experiments

4.1. Selection, Design, and Verification

4.1.1. System Overview - N² Chart



In the rover, power, thermal, data acquisition, communications, and mobility are the central systems that will either help or maintain the instruments. Power provides power for all the scientific instruments and allows them to do their roles on the rover until it is out or the mission is done. Thermal maintained the instruments as instruments could stop operating or could be damaged from the cold on Vesta. Communication is needed for cameras and data

acquisition. Communications allowed data to be retrieved from millions of miles away. With data acquisition, the current temperature of the rover, the readings of the ultraviolet spectrometer, and the images of the microscopic imager can be monitored from a computer. The mobility system is what will move the rover to the site and for pickup. Without the mobility system, the rover will have no way to proceed.

4.1.2. Subsystem Overview

In the rover, four science instruments are nested in the rover that will work together to achieve a goal, the collection of a Vesta's sample. The rock abrasion tool, microscopic imager, the ultraviolet spectrometer, and the radioisotope heater unit are the four science instruments on the rover. The rock abrasion tool generates 11 watts ("Rock Abrasion Tool"). The primary purpose of the rock abrasion tool is to drill areas of the three sites the rover will visit to find a suitable sample. The rock abrasion tool will take about two hours to dig volcanic rocks ("Rock Abrasion Tool"). After the rock abrasion tool drilled an area from either of the three sites, ultraviolet light will shine on the drilled area. The ultraviolet spectrometer will detect water particles through the ultraviolet light. Since water particles are blue, if the color blue is absent from the pierced area, the ultraviolet spectrometer will sense it ("Voyager - Spacecraft - Ultraviolet Spectrometer (UVS)"). The ultraviolet spectrometer generates 100 volts to 240 volts (USA, Hach). The microscopic imager will provide a close-up view of the drilled area. The close-up view of the drilled area is crucial as it reveals if water did exist in the area. The microscopic imager generates between 2.5 to 4.3 watts ("Microscopic Imager (MI)"). Lastly, the radioisotope heater unit provides the heating for

the rover through radioactive decay. It generates just 1 watt.

4.1.3. Precision of Instrumentation, Repeatability of Measurement, and recovery system

The ultraviolet spectrometer is the only data-gathering instrument in this rover. It has an absorbance precision of ± 0.005 ("Specifying Accuracy and Precision Criteria for Ultraviolet Spectrometers"). Because of this precision, it is essential to acknowledge that there will be flaws in the data that the ultraviolet spectrometer will be presented back to Earth. The precision value will affect the absorbance of it. This means that the amount of light that is being absorbed can change the color. Since the ultraviolet spectrometer detects any missing color, if the absorbance accuracy is being affected here, then the assumption that the color blue is missing is not enough to declare that there is water in the rocks. The ultraviolet spectrometer may not absorb all the light to conclude that decision.

The data received from the ultraviolet spectrometer will be sent back to the spacecraft and then to Earth. Because there are cameras on the rover and the microscopic imager, the camera feeds will also be sent back to Earth. Additionally, a sample will return to Earth.

4.1.4. Validation of Verification Plan

There are four instruments on the rover: the rock abrasion tool, the ultraviolet spectrometer, microscopic imager, and the radioisotope heater unit. To determine if the rock abrasion tool is working, since it is drilling and there will be cameras on the rover, seeing that it is not digging is enough to determine that it is not working. Since the microscopic imager has a camera in it, the fact that it is not displaying will tell whether or not it is working too. Since the ultraviolet spectrometer needs light to detect the colors, if it returns no color as its data, then that is enough to validate if it is working or not. An

item can't possess all the elements. Having a thermometer in the rover is enough to confirm if the radioisotope heater unit is working or not. The temperature in the rover should be constant or moving up. It should not go down. The point of the radioisotope heater unit is to keep the rover warm. If it goes out, then the temperature will go down as a result of that.

4.1.5. FMEA and Risk Mitigation

Because of the rocky, bumpy terrain of Vesta, the design can pose issues during the mission. If the rover's weight is not distributed equally, the rover could easily tip over because the bump can cause the rover to move to the sides. If the rover falls over, the objective of collecting a sample from Vesta will never be accomplished, and the whole mission will be aborted. Checking if the design of the rover will distribute the weight equally will mitigate this issue. Vesta's cold temperature also poses a problem with the scientific instruments. Instruments in general have a temperature range in which they can safely operate in. Vesta has an extreme temperature range. Vesta's average temperature is around negative 200 Fahrenheit (Dubar). As such, the scientific instruments may not possess the ability to operate in very chilly weather. The instruments could stop working because it can't handle the cold temperature. If the scientific instruments stop functioning, the mission can't be completed. If the ultraviolet spectrometer is gone, then there is no possibility of knowing if there is water in Vesta. If the rock abrasion tool is gone, then there is no drilling activity for the microscopic imager to see up close of the drilled area. However, in the Mars mission conducted by NASA, they too had to deal with the issue of the coldness. Mars gets as low as negative 140 Fahrenheit ("The Rover's Temperature Controls"). In order to combat the cold, incorporating whatever NASA did to protect their instruments from Mars at night is the team's best interest to protect the scientific instruments from Vesta. Lastly, the loss of communication is a problem for the team. The distance between Vesta and Earth is 156 million miles. The rover's communication

instrument mustn't get out of range. Doing so will prevent the camera feeds and data from the scientific instruments to not be sent to Earth. Losing those two can make the mission challenging to accomplish. Without the camera feed, the point of view of the rover will be gone, and it will be hard to see how the mission is going. Losing the data will cause difficulty as the ultraviolet spectrometer has a precision value in it. When the sample comes back on Earth, the data is used to determine the accuracy of the sample, whether or not there is water in it. Without it, the accuracy can not be determined, and the sample may not even hold water. To mitigate this, the rover must not be out of range of the spacecraft, and the spacecraft must be in the range where the rover will be visiting the three craters.

4.1.6. Performance Characteristics

Proving that the entire system will operate under Vesta's temperature is much easier to show than to prove that individual systems will work its temperature. From past NASA's missions, they too faced a similar situation just like Vesta. In their mission on Mars, Mars reached down to negative 104 Fahrenheit at night ("Rover Temperature Controls"). Mars at night is reaching down to the triple digits, just like Vesta. Because of this situation and the mission success, incorporating what NASA did on Mars to heat their rover can help warm the rover on Vesta. NASA used radioisotope heater units, gold paint, and aerogel to heat their rover. By incorporating these, the rover on Vesta can operate under Vesta's temperature based on past missions and its similarity.

4.2. Science Value

4.2.1. Science Payload Objectives

The team chose Vesta as an optimal landing site to collect data to better understand what it means for an asteroid to be considered a protoplanet, as well as how asteroids were able to hold water in the past.

Vesta is the best candidate for this research as some of its more prominent craters may contain samples left from past impacts, which may include water or other valuable residue. Similarly, Vesta's surface contains many rocks that can be collected from areas where there are high levels of water evaporation. By collecting this data, Vesta's rocks will help understand what a protoplanet is and help understand how other asteroids and planets were able to hold water despite water evaporating in the air.

4.2.2. Creativity/Originality and Uniqueness/Significance

The landing spot for the rover will be within proximity of the crater, known as Tarpeia. It has one of the largest diameters on Vesta, and due to its "freshness," it is an ample site for uncovering various minerals. It contains a slope, so to explore it, it will need a wheel system with sufficient traction to descend at a steady rate. It is also important to note that within this crater are various small craters close to one another, so an axle with a maximum turn radius would be more appealing in case any last-minute turns are necessary. The temperature on Vesta, in general, is also freezing; therefore, the rover itself would need a form of insulation so that its components do not freeze when not in motion or when in action. Lastly, a compartment to collect the samples would be required assuming the temperature conditions can be matched as to keep the sample within the same element conditions

4.2.3. Payload Success Criteria

For the mission to be considered a success, an effort must be made to retrieve a sample that can be preserved at its freshest point. To do this, the team would like to use one of the instruments that were

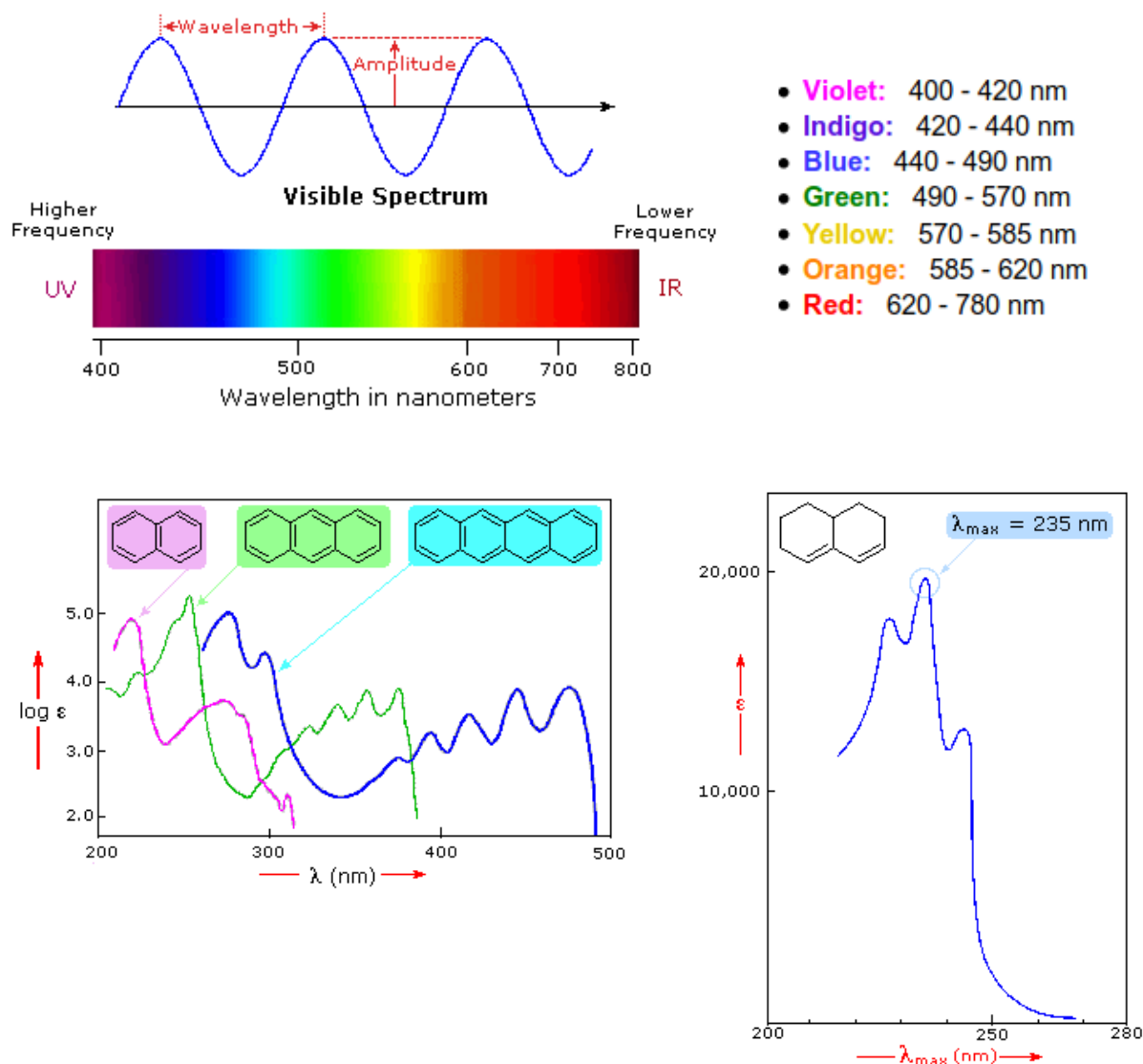
used on a previous mission to mars known as RAT. This tool will dig into a specimen that is essentially encapsulated with the hopes that it has been preserved, given the low temperatures on Vesta. The ultraviolet spectrometer that will be incorporated will be calibrated to detect particles of water. It also an understanding that due to the cold temperatures on Vesta we must add some kind of sealer to our probe in order to keep the parts from freezing without compromising mobility. Before the sample is collected, a microscopic imager will be used to make a quick evaluation if the sample is worth retrieving.

4.2.4. Describe Experimental Logic, approach, and method of investigation

4.2.5. Describe Testing and Measurements, including variables and controls

To calibrate the instruments, running all of them the moment it lands on Vesta is the goal. By running all of them, it ensures that every instrument is working and starts getting the heat in the rover. Specifically, testing the ultraviolet spectrometer and the microscopic imager is required. Testing the ultraviolet spectrometer through light is enough to calibrate the instrument. By checking the light, it will give the team an idea of the accuracy of it due to its precision value. The microscopic imager needs to be tested as a camera is in it. The camera will require some calibrating in it. All of these will be based on Vesta's temperature. While the temperature, in general, is not constant, Vesta's temperature is not shifting drastically compared to the temperature inside of the rover. Once all of these are tested, the testing of the wheels is necessary to ensure that the mobility system works. Driving the rover around is the plan for the calibration of the mobility system.

4.2.6. Show expected data & analyze



The ultraviolet spectrometer has the ability to detect elements based on the missing color. Since it will return data similar to the ones shown above, it has demonstrated shades of indigo, green, and blue. Since the colors are based on ROY G BIV, it is safe to assume that the colors that are not showing on the data, the item contains those elements. If the color blue was missing on that data, the assumption of that particular element is presented in that item. However, because of the precision value that is attached to it, the spectrometer may not absorb all the light and colors despite it missing could have the possibility of not even being in the item.

5. Safety

5.1. Personnel Safety

5.1.1. Safety Officer

Ryan is the safety officer on this team. Ryan's responsibilities include identifying the hazards the team and the rover could face during the mission. He identified two types of risks: dangers from Earth and hazards from Vesta. To identify the individual hazards first, the possible activities had to be listed first. Understanding Vesta's environment was needed to determine the danger the teammates could face when it comes to testing the rover on a similar environment like Vesta on Earth. In addition, solutions or migrations were needed to be research in order to protect the team and the rover from the mission.

5.1.2. List of Personnel Hazards

The possible activities the team will be engaging are assembling the instruments on the rover, visiting locations that hold similar conditions like Vesta, testing the rover, the launch itself, and handling the collected sample from Vesta.

5.1.3 Hazard Mitigation

Assembling the rover holds danger because of electrical wires, and the assembling itself. To migrate the electrical wires issue, the team must not turn on the parts as well as wear work boots and not touch any metals during the handling of the wires. With the assembling, teammates must not work alone as they could hurt themselves in the process. Working

together helps minimize the damage they could do to themselves and the parts. With Vesta being a cold and bumpy asteroid, the team must find locations that bear these similarities. To decrease the chance of frostbite, the team must wear clothes designed for the cold to stay warm and test the rover in intervals, so the team is not outside the whole time when it comes to testing. Testing the rover at some point, will reach a destination that is unsafe for the team to rescue it. Also, the rover may malfunction, and the results of that or the individual instruments are unknown. Thus, the team may not recover the rover by hand and must be at least $\frac{1}{4}$ of a mile away from the rover in case of the explosion of the rover or the individual parts. With the launch, the whole team must be in the control room to ensure that no one is near or outside the start when it happens. Lastly, the collected sample must not be touched by teammates as the uncertainty of what the samples could do to the human body is unknown.

5.2. Lander/Payload Safety

5.2.1. Environmental Hazards

The most hazardous issue Vesta holds is how cold it could be. Vesta's average temperature is around negative 200 Fahrenheit (Dunbar). The rover and the instruments must be able to operate and withstand around that temperature.

5.2.2. Hazard Mitigation

To migrate the issue, the material of the rover is made out of aluminum as aluminum can withstand against very low temperatures ("Properties of Aluminum Alloys at Cryogenic and Elevated Temperatures"). So, the wheels of the rover must be

made out of aluminum also. The instruments must need some heating around it to continue operating in the cold.

6. Activity Plan

6.1. Budget

The budget cap for this mission is forty-five million dollars. To start, the overall cost of this project will have a 30% margin added to it. Manufacturing will have a 50% margin added to it, and facilities and administrative costs will have a 10% margin added to it as well. The cost of the rock abrasion tool is \$1,500,000, and the radioisotope heater unit is \$117,594 each. Since five are needed on the rover, it would cost \$587,970. The cameras cost \$7,212 each. There will be four cameras so the price will cost \$28,848. However, the price of the microscopic imager and the ultraviolet spectrometer could not be found, so their prices would be based on similar instruments. The ultraviolet spectrometer cost around \$9,488 ("DR 6000™ UV VIS Spectrophotometer"). The microscopic imager cost around \$650 ("Zeiss Microscope Objective Planapo 40x with Correction."). The overall cost of the scientific instruments is \$2,126,916. The project will take around five years. Since there is 8 in the team and each salary coupled with a bonus is \$102,400, the total cost of all five years is \$4,096,000. The manufacturing cost is \$2,215.88. The duration of the mission when it takes place on Earth is five days in Cape Canaveral, Florida. The flight will cost \$8824, and the hotel will cost 1024. Food will cost \$528 and each member will have a spending amount of \$576. The team will need to rent a car so it will cost around \$360. The total cost of this is \$11,312. The margin cost of the manufacturing is \$4431.76. The margin cost of the facilities and administrative costs for the 5 years of working on this project is \$4,551,111.11. The margin cost of the trip is \$12,568.89. Overall the total cost with the margin is \$9,564,325.36.

6.2. Schedule

The team is scheduled to have the rover over on Vesta's land and collect data to fulfill the mission criteria. The science team will research Vesta and Eros to determine which asteroids will satisfy the mission criteria. They will also research scientific instruments and the asteroid's environment they choose to help with the design team with assessing the design of the rover. The design team will investigate the design for the rover, including looking at past missions' models and the asteroid's environment in mind. Researching the design of the rover also includes looking at the descent system of the rover. After the design is chosen, the risks and budget will be discussed in time for the PDR. The model of the rover reviewed for the CDR to ensure that the design meets the requirements stated by the mission concept and ensure that it will also support the mission full-scale. After the completion of the PDR and CDR, Phase C will begin with manufacturing the rover and testing the individual systems. The testing will be conducted in an environment similar to Vesta's. After Phase C is done, Phase D begins with integrating the scientific instruments and the descent system. After Phase D ends, begins the preparation for launching the team's rover to Vesta. The team will get together on October 15th in Cape Canaveral in Florida for the meeting and the launch. The rover will be launched on October 15th. The team will monitor the rover and the mission until October 17th, in which the team will depart.

6.3. Outreach Summary

For the outreach program, setting up science days in which future scientists/engineers are giving small kits to assemble and then given a flat terrain that they then implement their design for retrieval purposes is a possibility. Once that's concluded, they can present a powerpoint that shows how this small task is related to a much bigger mission involving outer space. Space, in general, is exciting because of the mystery of not knowing what's out there.

6.4. Program Management Approach

The team will approach the project known as “Agile Project Management.” Agile Project Management builds upon flexibility. As the project evolved, sections of the PDR are bound to develop. Agile Project Management allows the team to respond to change.

7. Conclusion

Overall to come to a satisfactory conclusion for this project, evaluating the terrain that the rover would be exploring was needed. From there, a design was formulated that would be the most accommodating based on the restrictions provided. Another resource that was used that helped with the design was the information available from previous missions. From the research that was gathered, this was the best possible solution that would result in success in determining another source of water that could lead to another habitable planet. The next milestone would be to research another possible habitable planet, if one is determined to exist, once the origins of water are determined.

Bibliography

- Dunbar, Brian. "Vesta Likely Cold and Dark Enough for Ice." *NASA*, NASA, www.nasa.gov/mission_pages/dawn/news/dawn20120125.html.
- "Properties of Aluminum Alloys at Cryogenic and Elevated Temperatures." *Total Materia*, Aug. 2001, totalmateria.com/article23.htm.
- "Rock Abrasion Tool." *Honeybee Robotics*, honeybeerobotics.com/portfolio/rock-abrasion-tool/.
- "Microscopic Imager (MI)." *NASA*, NASA, mars.nasa.gov/mer/mission/instruments/mi/.
- USA, Hach. "DR6000™ UV VIS Spectrophotometer with RFID Technology." *Hach*, www.hach.com/dr6000-uv-vis-spectrophotometer-with-rfid-technology/product-details?id=10239244800.
- "Voyager - Spacecraft - Ultraviolet Spectrometer (UVS)." *NASA*, NASA, voyager.jpl.nasa.gov/mission/spacecraft/instruments/uvv/.
- "The Rover's Temperature Controls." *NASA*, NASA, mars.nasa.gov/mer/mission/rover/temperature/.
- "Specifying Accuracy and Precision Criteria for Ultraviolet Spectrometers." *Spectroscopy Europe/Asia*, John Wiley & Sons Ltd and IM Publications LLP, <https://www.spectroscopyeurope.com/quality/specifying-accuracy-and-precision-criteria-ultraviolet-spectrometers>.
- "DR 6000™ UV VIS Spectrophotometer." *Thomas Scientific - Laboratory Supplies*, *Laboratory Equipment, Laboratory Chemicals, and Laboratory Safety*,

https://www.thomassci.com/Instruments/Spectrophotometers/_/DR-6000-UV-VIS

-Spectrophotometer?q=Uv Vis Spectrophotometer.

“Zeiss Microscope Objective Planapo 40x with Correction.” *Zeiss Microscope Objective Planapo 40x with Correction | For Sale | Labx Ad LV40411968*,

<https://www.labx.com/item/zeiss-microscope-objective-planapo-40x-with-correction/LV40411968>.